

MUSFIQUR RAHMAN

AI/ML Engineer & Researcher | LLMs, NLP & Applied Machine Learning

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SUMMARY

PhD candidate and AI/ML Engineer with 8+ years of combined research and industry experience building LLM-powered systems, evaluation frameworks, and production ML pipelines. Strong Python engineering discipline with a track record of shipping reliable, tested, end-to-end AI systems. Deep expertise in NLP, LLMs, RAG, adversarial robustness, and model evaluation. Published at ICSE, MSR, EASE, and EMSE with 100+ citations. Daily user of Claude, ChatGPT, and Gemini for coding, writing, and research workflows.

CORE TECHNICAL SKILLS

Languages: Python (expert), Java, C/C++, SQL, R, Bash

AI/ML: LLMs, NLP, Transformers, Deep Learning, PyTorch, CatBoost, Scikit-learn, HuggingFace

LLM Stack: Prompt Engineering, RAG, Embeddings, LLM-as-Judge, Agentic Workflows, Model Evaluation

Data: Pandas, NumPy, PostgreSQL, BigQuery, large-scale data pipelines, ETL

DevOps/Infrastructure: Docker, Git, GitHub Actions (CI/CD), GCP, AWS, Azure ML, Streamlit

Statistics: Hypothesis testing, survival analysis, causal inference, bootstrap methods, experimental design

Visualization: Matplotlib, Seaborn, Streamlit, Power BI, Tableau

EXPERIENCE

PhD Researcher — LLMs for Software Engineering

May 2022 – Present

DAS Lab, Concordia University (Advisor: Prof. Emad Shihab), Montréal, QC

Research focused on LLM code generation, detection, evaluation, and mechanistic interpretability. All projects involve production-level engineering alongside empirical research. NSERC SE4AI CREATE Program Trainee.

Project 1: SIEVE — Scalable LLM-Generated Code Detection System

Sep 2025 – Apr 2026 | Live: sieve-tool.streamlit.app | github.com/mrsumitbd/sieve

- **About:** An end-to-end system for detecting LLM-generated code at scale, designed to help researchers and practitioners identify whether code was written by a human or generated by an LLM. Includes a GitHub-based data discovery and ingestion pipeline, multi-language code extraction, deduplication, transformer-based classification, and a live production deployment.
- **My Role:** Sole engineer and researcher. Designed the system architecture, curated the 114K-sample dataset spanning 9 LLMs, trained the CodeBERT-based classifier (F1=0.9478, AUROC=0.9902), built the CI/CD pipeline, and deployed the live tool. Built the HuggingFace model and dataset releases.
- **Tech Stack:** Python, PyTorch, HuggingFace Transformers, CodeBERT, Scikit-learn, Streamlit, Docker, GitHub Actions (CI/CD), PostgreSQL, GCP, REST APIs, pytest (273 tests, 80% coverage)

Project 2: OpenClassGen — Large-Scale Python Class Benchmark Dataset

Apr – Nov 2025 | Published: EASE 2026 | DOI: 10.1145/3816483.3816547 | HuggingFace: [mrahman2025/OpenClassGen](https://huggingface.co/mrahman2025/OpenClassGen)

- **About:** A benchmark dataset of 324,843 real-world Python classes extracted from 2,970 GitHub repositories, designed to evaluate LLM code generation under realistic conditions. Includes 27 structural and stylometric quality metrics per class. Released publicly for use by the research community.
- **My Role:** Sole engineer and researcher. Designed and built the full data pipeline: GitHub API-based repository discovery, Python AST extraction, deduplication, metric computation, quality validation, and dataset packaging for HuggingFace release.
- **Tech Stack:** Python, GitHub REST API, AST parsing (tree-sitter), Pandas, NumPy, HuggingFace Datasets, PostgreSQL, GitHub Actions

Project 3: RealClassEval — LLM Evaluation Framework for Real-World Code Generation

2025 – Present | Under Revision: EMSE | arXiv: 2510.26130

- **About:** An evaluation framework built on OpenClassGen to benchmark LLM code generation under real-world conditions. Evaluated GPT-o4-mini, Claude-4-Sonnet, and Qwen-3-Coder across four quality dimensions: syntactic (BLEU, ROUGE-L), semantic (CodeBERTScore), structural (Tree Similarity Edit Distance), and functional correctness (pass rate via Pynguin test generation). Surfaced a 50–65% gap between synthetic benchmark performance and real-world conditions across 7 LLMs.
- **My Role:** Designed and implemented the evaluation framework, including RAG-based context enrichment strategies (zero context through dependency-aware retrieval), metric computation pipelines, and statistical analysis. Ran experiments on Compute Canada HPC.
- **Tech Stack:** Python, OpenAI API, Anthropic API, HuggingFace, CodeBERT, Pynguin, tree-sitter, Pandas, SciPy, Compute Canada (A100-40GB GPUs, SLURM)

Project 4: Automatic Detection of LLM-Generated Code — Adversarial Robustness Study

2023 – Present | Under Revision: EMSE | arXiv: 2409.01382

- **About:** A comprehensive study of LLM-generated code detection across 4 LLMs (GPT-3.5, GPT-OSS, Claude 3 Haiku, Claude Haiku 4.5) and two granularities (function and class level). Includes an adversarial robustness experiment testing classifier degradation under post-hoc comment stripping and few-shot regeneration attacks. Dataset: 14,485 functions + 11,913 classes from CodeSearchNet.
- **My Role:** Lead researcher and engineer. Designed the feature engineering pipeline (44 structural and stylistic features), built and trained CatBoost classifiers, designed and executed the adversarial attack conditions, and conducted statistical analysis with bootstrap confidence intervals (1000 iterations).
- **Tech Stack:** Python, CatBoost, Scikit-learn, AutoSpearman, OpenAI API, Anthropic API, Pandas, SciPy, Matplotlib, Compute Canada HPC

Project 5: AI Code Retention in Open Source — Survival Analysis Study

Jan 2026 – Present | Under Revision: TOSEM

- **About:** A large-scale study of how AI-attributed code (from pair-programmer tools, batch generators, and autonomous agents) survives in production open-source repositories over time. Dataset: 156 repositories, 222,299 line-level + 45,857 hunk-level units. Applies survival analysis and causal inference to compare retention rates across AI tool categories.
- **My Role:** Lead researcher. Designed the 3-tier taxonomy of AI tool types, built the data collection pipeline using GitHub API, implemented survival analysis (Kaplan-Meier, Cox PH, stratified analysis), and conducted a matched-task analysis (269 pairs, 24 repos) to control for confounding.
- **Tech Stack:** Python, GitHub REST API, lifelines (survival analysis), SciPy, Pandas, tree-sitter (17 languages), R (Cox PH), Matplotlib

Project 6: LLM-as-Judge Evaluation Panel — Intent Validation Pipeline

2025 – Present

- **About:** A multi-LLM agentic evaluation pipeline designed to validate intent classification labels at scale (n=385 samples). Three LLM judges run in parallel, with a priority-rule arbitration layer that reconciles disagreements against a ground-truth human annotation scheme.
- **My Role:** Designed and implemented the full pipeline: judge prompt design, async API orchestration across three providers, structured output parsing, priority-rule adjudication logic, and evaluation against human annotation.
- **Tech Stack:** Python, Together.ai API (GPT-OSS-120B, Qwen3.5-9B, Llama-3.3-70B), asyncio, Pandas, JSON schema validation, GitHub API (live data fetching)

Project 7: Mechanistic Interpretability of Code LLMs

2025 – Present

- **About:** Empirical study tracing how transformer-based code LLMs (Qwen2.5-Coder, DeepSeek-Coder) internally retrieve and apply knowledge during library-selection decisions. Uses attention head analysis, logit lens, and activation patching to localize decisions to specific layers and heads.
- **My Role:** Sole researcher. Designed the experimental protocol, built the activation patching and logit lens analysis pipelines, and identified specific attention heads (L20-H11, L21-H0 in Qwen; L10-H13 in DeepSeek) responsible for library-selection decisions.
- **Tech Stack:** Python, PyTorch, HuggingFace Transformers, TransformerLens, Compute Canada (A100-40GB GPUs, SLURM)

Enterprise Solutions Architect & AI Systems Evaluator

Nov 2025 – Jul 2026

General Assembly, Remote

Evaluated and guided AI systems for enterprise clients and GA's internal AI infrastructure.

Project 1: GAIA Evaluation — AI Agent System QA

- **About:** GAIA is GA's proprietary AI agent system for automated course content generation from learning objectives and student personas. My role involved systematic quality assurance, evaluation protocol design, and failure mode analysis.
- **My Role:** Designed and executed evaluation protocols for agentic multi-step workflows; assessed output quality for pedagogical soundness, factual accuracy, and structural coherence; documented failure modes for engineering stakeholders.
- **Tech Stack:** Python, prompt engineering, structured evaluation rubrics, internal GA systems

Project 2: Enterprise AI Solution Design

- **About:** Scoped and designed custom AI learning solutions for enterprise clients including Adobe, Diageo, Rightmove, and Northwest Education & Training using an L0–L4 content customization framework.
- **My Role:** Client-facing technical lead. Translated ambiguous business requirements into scoped technical specifications; recommended AI tool adoption strategies and content architectures appropriate to client needs.

- **Tech Stack:** GA internal systems, LLM-powered content tools, Microsoft 365

Subject Matter Expert & Lead Educator, Data and AI

Feb 2020 – Nov 2025

General Assembly, Remote

Designed and delivered curriculum across multiple data science and AI bootcamp programs; trained 1,000+ professionals across 50+ cohorts globally.

Project 1: Data Science Immersive Curriculum

- **About:** 12-week immersive bootcamp covering Python programming, statistics, machine learning, deep learning, and industry applications.
- **My Role:** Lead instructor and curriculum developer. Designed labs, projects, and assessments; mentored junior instructors; iterated content based on learner feedback.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, SQL, Jupyter, Git

Project 2: Data Analytics Bootcamp (Power BI & Tableau)

- **About:** Curriculum covering data analysis, visualization, and business intelligence for non-technical professionals.
- **My Role:** Lead instructor. Designed and delivered hands-on Power BI and Tableau content for business audiences.
- **Tech Stack:** Power BI, Tableau, SQL, Excel, Python

Project 3: GenAI Applications & Frameworks Course

- **About:** Course covering practical GenAI application development including prompt engineering, RAG, and agentic workflows for enterprise audiences.
- **My Role:** Subject matter expert and curriculum developer. Designed hands-on labs using OpenAI, Anthropic, and HuggingFace APIs.
- **Tech Stack:** Python, OpenAI API, Anthropic API, HuggingFace, LangChain concepts, Jupyter

Data Scientist (Part-time)

May 2023 – Jan 2024

SGS, Remote

Applied time-series statistical modeling for financial forecasting in a consulting context.

Project: Financial Time-Series Forecasting

- **About:** Developed and deployed time-series forecasting models to support financial planning and business decision-making.
- **My Role:** Data scientist. Built, validated, and documented forecasting models; delivered insights to business stakeholders.
- **Tech Stack:** Python, statsmodels (ARIMA, SARIMA, SARIMAX), Facebook Prophet, Pandas, Matplotlib, Azure ML

Artificial Intelligence Engineer

Feb 2018 – Feb 2020

Pentavere (Health Tech), Toronto, ON

Founding AI engineer. Built and deployed production clinical NLP systems processing millions of unstructured health records in a regulated healthcare environment. Collaborated closely with clinical stakeholders to define accuracy requirements, ensure regulatory compliance, and validate model outputs against clinical ground truth.

Project 1: Clinical Named Entity Recognition (NER) — Medication, Dosage & Clinical Artifact Extraction

- **About:** Production NER system for extracting structured clinical entities from unstructured physician notes and medical records at scale, including medications, dosages, frequencies, routes of administration, and other clinical artifacts (lab values, diagnoses, procedures).
- **My Role:** Designed and trained the NER pipeline, including annotation schema definition, model training and evaluation, and integration into the production data extraction pipeline. Collaborated with clinical experts to validate entity boundaries and label quality.
- **Tech Stack:** Python, spaCy, NLTK, Scikit-learn, custom annotation pipelines, PostgreSQL, AWS, Docker

Project 2: Smoking Status Classifier — Binary Classification from Doctor's Notes

- **About:** A text classification model to automatically identify and classify a patient's smoking status (current smoker, former smoker, non-smoker, unknown) from unstructured physician notes. Used as a structured feature in downstream clinical risk models.
- **My Role:** Built the full classification pipeline: data preprocessing, feature engineering from clinical text, model training and evaluation, and integration into the production system. Validated against clinician-reviewed labels.
- **Tech Stack:** Python, spaCy, NLTK, Scikit-learn, TF-IDF, Pandas, PostgreSQL, AWS

Project 3: OCR Pipeline — Handwritten Clinical Notes Digitization

- **About:** An OCR-based pipeline to convert handwritten physician notes and clinical documents into machine-

readable digital text, enabling downstream NLP processing on historically paper-based records.

- **My Role:** Built and integrated the OCR pipeline into the clinical data ingestion workflow, including preprocessing (image normalization, noise reduction), OCR model integration, post-processing text cleanup, and quality validation against ground-truth transcriptions.
- **Tech Stack:** Python, Tesseract OCR, OpenCV, PIL, image preprocessing libraries, PostgreSQL, AWS

Project 4: ICD-10 Classification — Automated Medical Coding from Doctor’s Notes

- **About:** A multi-class classification system for automatically assigning ICD-10 diagnostic codes to physician notes, enabling automated medical billing and clinical coding at scale. ICD-10 has thousands of possible codes, requiring a hierarchical or multi-label classification approach.
- **My Role:** Designed and trained the classification model, including text feature extraction from clinical notes, handling of class imbalance across thousands of ICD-10 categories, and evaluation against professional medical coders as ground truth.
- **Tech Stack:** Python, spaCy, NLTK, Scikit-learn, TF-IDF, deep learning (TensorFlow), PostgreSQL, AWS, Docker

Graduate Research Assistant

Sep 2015 – Apr 2018

CESEL Lab, Concordia University (Advisor: Prof. Peter C. Rigby), Montréal, QC

M.Sc. research on code naturalness and statistical machine translation between English and source code.

Project 1: Natural Software Revisited

Published: ICSE 2019 (Acceptance Rate: 20%)

- **About:** Large-scale empirical study analyzing the predictability of programming languages using n-gram language models, revisiting foundational claims about code naturalness.
- **My Role:** Lead researcher. Built language models for 8 programming languages, conducted statistical analysis, and authored the publication.
- **Tech Stack:** Python, NLTK, SRILM, MITLM, R, PostgreSQL

Project 2: Cleaning StackOverflow for Machine Translation

Published: MSR 2019 Data Showcase

- **About:** Built and released a cleaned parallel corpus of English-to-code pairs from StackOverflow for statistical machine translation research.
- **My Role:** Designed and implemented the data cleaning and parallel corpus construction pipeline; released the dataset for the research community.
- **Tech Stack:** Python, NLTK, Moses (SMT), PostgreSQL, Bash scripting

EDUCATION

PhD in Software Engineering

May 2022 – Sep 2026 (Expected)

Concordia University, Montréal, QC

GPA: 4.30/4.30

Thesis: *Large Language Models in Coding: Generation, Detection, and Repair*

M.Sc. in Computer Science

Sep 2015 – Apr 2018

Concordia University, Montréal, QC

GPA: 3.90/4.30

B.Sc. in Computer Science and Engineering

Jan 2011 – Dec 2014

East West University, Dhaka, Bangladesh

GPA: 3.62/4.00

SELECTED PUBLICATIONS

- **M. Rahman** et al. “OpenClassGen: A Large-Scale Corpus of Real-World Python Classes for LLM Research.” *EASE 2026*. DOI: 10.1145/3816483.3816547.
 - **M. Rahman** et al. “Beyond Synthetic Benchmarks: Evaluating LLM Performance on Real-World Class-Level Code Generation.” arXiv:2510.26130. [Under Revision – EMSE]
 - **M. Rahman** et al. “Automatic Detection of LLM-Generated Code.” arXiv:2409.01382. [Under Revision – EMSE]
 - **M. Rahman** et al. “The Impact of Environment Configurations on the Stability of AI-Enabled Systems.” *EASE 2025*.
 - **M. Rahman**, D. Palani, P. C. Rigby. “Natural Software Revisited.” *ICSE 2019*.
 - **M. Rahman**, P. C. Rigby. “Cleaning StackOverflow for Machine Translation.” *MSR 2019*.
- 100+ citations | [Google Scholar](#)

AWARDS & ACADEMIC SERVICE

- **Concordia University Doctoral Fellowship** (2022–2026) – Full PhD funding for research excellence
- **NSERC SE4AI CREATE Program Trainee**

- **Campaign for Concordia Graduate Award** (2022)
- **Academic Service:** PC member/reviewer for ICSME 2026, ICSE-SRC 2026, MSR 2024/2026, TechDebt 2026, CAIN 2026, TOSEM, EMSE